

ActInf Textbook Group ~ Cohort 6 ~ Session 2 (Chapter 1, Part 1) ~ 2/5/2024

TextbookGroup/ParrPezzuloFriston2022/Cohort_6 | Cohort 6 ~ Session 2 (Chapter 1, Part 1) ~ 2/5/2024 | 56:12

all right thank you everyone today is
February 5th 2024 and Oli will be
facilitating this discussion on chapter
one for cohort 6 so thanks
Oli okay uh hello everyone thanks for
joining for our first discussion for
chapter one uh well um as you might have
read in in the content and uh also for
the introduction in chapter one uh this
is sorry this chapter gives a kind of um
overall picture of uh what kind of
theory active inference is uh and
specifically uh what kind of approach uh
the textbook is going to take explaining
different aspects of um active inference
Theory uh well one of the main themes of
uh this chapter and U also the overall
organizational scheme of the textbook is
uh the difference
between two kind of approach to active
inference literature namely uh the high
road and the low road so basically uh
for high road uh the prior assumption is
the existence of quote unquote things
and then explaining from that um
seemingly obvious assumption into what
uh what requirements that quote unquote
thing uh should have in order for it to
have U many various uh aspects such as
agency uh as a kind of decision maker
and so on so basically uh it's not
something specific to cognitive agents
but uh the whole framework can also be
applied to many other uh uh many other
agents uh as long as it can be described

in terms of its internal and external Dynamics so uh that's one approach to active inference the other one uh which uh is usually called Low Road in the literature is beginning from uh some of the concepts developed uh from back in the 19th century uh for example by people such as Helmholtz uh in which they kind of they it tries to generalize uh the Bayesian reasoning uh into um onto un developing uh the notion of free energy and specifically variational free energy uh for which we can um we can formulate uh the way that that the organisms or agents can perceive the world and also uh the related variable or related so-called parameter uh so for uh for perception we need a kind of lower boundary which is the AAL free energy and uh for um for control or for decision making uh we need uh another uh variable or parameter namely expected free energy so by describing these two parameters as kind of lower bound of certainty we can uh again formulate how uh organisms or cognitive agents can engage in both perception and also action so basically when we say active inference it's uh it implies a kind of integration of these two um these two aspects of perception and action so they're not uh essentially distinct from each other uh both of them uh both of them kind of um follow a similar trajectory or similar mathematical technology but uh in somewhat mirror or

uh reversed way so um I hope I uh I
could give a kind of overall meaning of
what high road and low road uh mean but
obviously we'll uh we'll get to see much
more detail of both of these approaches
so the uh the chapter two deals with um
the introduction of low road approach
and then beginning from chapter 3 we'll
see how high road approach differs from
uh the low road one so um if anyone has
any thought or comments or question on
these two concepts or we can go to the
question section and

see uh and discuss some of the questions
positive

there

okay so maybe we can open the question
section

or by the way

yeah no on that question um so in the in
the chapter it talks about the fact that
organisms or just agents can not the
motivation is to reduce surprise but
they may not able to act on surprise
directly but instead on what's called
proxies could you and they see that's a
variation the free end variation free
energy could you clarify what we mean by
that or just give some examples of that
sure so yeah the concept of surprise uh
actually is a little bit different than
uh the psychological conception of
surprise uh that we're familiar with in
everyday parlance but uh basically what
we mean in active inference by surprise
is its uh statistic in its a statistical
sense so uh it's surprisal as a
functional uh and uh yes so variational

free energy is a kind of upper bound of this function or this surprisal this parameter θ in which θ so the the agent π when it engages in the act perception it tries to minimize this upper bound or at least θ to somehow optimize its error θ so that the θ KL Divergence of its error wouldn't exceed this θ bound to this variational free energy so basically θ by specifying this θ upper bound or or this bound as an optimization parameter θ we can deal with approximate bayesian inference instead of exact bayesian inference so we don't need to exactly or precisely specify θ the Divergence of θ the perception from θ the input data θ instead we can specify a kind of θ bound to to approximately θ minimize that error in order θ for the perception to have some meaningful π to have any kind of meaning θ for us θ in terms of its action or any θ subsequent π subsequent message passing and so on θ so yes here θ variational free energy again is a parameter is an optimization parameter θ so θ the agent in active inference framework θ will try to somehow minimize this θ or actually optimize θ this parameter based on the Divergence more specifically the KL callback lower Divergence between the input data and its θ generative model we'll see how it works θ more precisely and More rigorously in chapter

two but uh free energy yes is the surprisal or the parameter uh the agent tries to minimize and variational free energy is kind of a get around uh so that instead of engaging in uh exact basian inference we just need to do the approximate basian inference which is much more tractable than the exact basian inference because as we know basian inference uh for U anything more than very very simple cases are extremely intractable I had a a quick question about that if that's okay yes sure um and by the way it's okay if we need to put a pin in this and um and perhaps address it in a later chapter as well I am I am interested in um not being super familiar with um with the that intractability problem do you know what it is that makes that straight up basian computation in intractable and and how does uh variational free energy get around that or should we save that for later

time okay so a preliminary answer would be uh so in order to do um an exact basian inference we need to account for every single parameter every single variable that uh that is relevant uh for U any kind of real real world situation uh so as we know uh when when we want when we want to uh want to model any any kind of real world situation uh basically the sheer number of variables we need to account uh are is extremely uh I

mean uh say for some very very simple cases in other cases we need to account for many many more variables so it makes uh the situation much more complicated uh so um and also we might not also know uh the exact Val for every parameter or even what parameters to account for right uh but uh when we want uh when uh as a way to get around this uh problem uh we might want to engage in a proximate basian inference in which uh instead of uh Computing the exact value of a prior we approximate it by doing a kind of margin alization on uh the variables we know the value of it of them and so it kind of gives uh much more um realistic way uh to compute those um those posteriors posteriors uh in any kind of situation so uh that's a much more tractable situation than U or basically we can say that Computing exact basian inference uh in uh most situations can be seen not only intractable but almost impossible um as I said we might not know the exact value of the parameters we want to model so uh yes that's uh one of the um essential issues uh of doing any kind of basing statistics basian inference and um almost always uh the the number here in the parameter in the denominator p_y is almost always marginalized in um real work situations so that marginalization over p_y uh instead of the exact value of p_y um is basically called approximate

basian

imprints fantastic it was very clear

thank you just to add like one more

point there because it's a great

question so like in the setting of

basian phog genetics X the hidden state

was the real relationship among species

unobserved and then Y is a huge data set

so it might be millions of lines or

something like that um the likelihood of

that data set being observed like Ali

said like if you knew that you'd kind of

have already known so not only is it a

massive data set but but it this

question seems to require you to know

what you already want to know and then

um the variational inference which is a

computational heris approach that's

existed long before active inference

like variational autoencoders and the

technique here is kind of like reducing

to a certain family of models that only

have a few knobs to fiddle with like I'm

going to fiddle with the mean and the

variance of this distribution and that's

it and then there are smoothly

incrementally optimizable paths for this

sorry for this kind of a problem that

you can take in time incrementally

versus this which is just like a massive

static problem that is intractable for

sometimes a number of reasons and so the

approach taken to the action perception

Loop in active inference just to a first

pass is to treat it like this kind of

structure fixed though you know asterisk

later developments but just at the core

level is to treat it like a structurally

fixed optimization
problem rather than this sort of static
number
crunch much appreciated thank
you um and also as a side note um
there's this fascinating pre-print uh
which was I think released um a couple
of weeks ago namely naturalizing
relevance realization so the concept of
relevance realization uh by John Veri is
uh very relevant here so
uh basically what they claim in that
paper is uh cognition is
fundamentally uh uncomputable in its in
uh very well- defined algorithmic sense
but uh predictive processing
uh via using a
probabilistic uh modeling instead of uh
using the exact values for parameters uh
they can actually provide a much more
realistic way uh to in order to model uh
the information processing nature of uh
any kind of uh biological organisms uh
obviously including the brains so I
highly recommend reading uh this paper
to I think yeah it is this one so yeah
so basically relevance
realization so what John Veri means by
relevance uh is exactly uh the
parameters we need to uh model or we
need to account for in order to
meaningfully model the behavior of the
organism uh which is
huge okay so any other comments
questions
or we can look at some of the
questions and the coda
page

all right so

uh uh we may dwell a little bit on the distinction between the low road and high road and as career uh opinions uh I mean before reading uh the book for uh those of you uh uh who are approaching this book for the first time uh what do you think uh the

advantages or disadvantages of using um each of these approaches uh to active influence literature based on uh this uh preliminary understanding of the distinction between these two approaches or which one would you find more meaningful I don't know more logical and so

on I I would say um I was struck by the way that um the high road provides this sort of like um great overview you know like a very elegant explanation that just begins with this simple fundamental predicament of self-organizing systems um so I find it very useful conceptually in that sense I was explaining trying to explain active inference to someone who's not familiar with it recently and uh I was kind of talking about some of these Concepts and he said uh yeah but that's you're you're using mentalizing Concepts right um and uh the a agentic

perspective um that's introduced by the high road model I think is interesting and and I'd like to um learn more about how this relates to this sort of um ontology like what the ontology behind it is because it seems to really um go against a lot of the sort of um

you know the approach uh the sort of
deterministic
um uh you know traditional approach
where there's this gap between kind of
agents and Consciousness and mentalizing
Concepts and then a world that's purely
determined like is there formal
causation because creatures have evolved
you know um that's a little mysterious
to me so that's that's what struck
me great thanks uh yeah uh that's a
great point I mean uh one of actually as
far as I know one of the main discussion
points of whole active inference
literature is exactly uh focused on uh
kind of unentangled uh what kind of
metaphors they use and uh making them
more uh precise uh so um I agree that
some of some of the metaphors used in
active inference can seem at first uh
kind of I don't know uh appealing to a
kind of panych
psychism kind of kind of approach or um
I don't know make them uh a bit uh more
anth
anthropocentric than it than it really
is uh but um it's important to
keep uh this point in mind that uh some
of the terms used in active inference
are used in their U very technical and
uh mathematically rigorous sense so for
instance uh there was a debate around
the use of the word sentience uh in one
of the papers which published u a few
months
ago so uh the way active inference
literature uses this word sentience is
actually uh specifically and rigorously

defined in some of the papers uh and it's not necessarily something that perfectly maps onto the everyday conception of of this word or even uh in the sense of sense of the this word that's used in some other relevant disciplines or other related areas uh but yes that's a great point and uh by the way

uh I almost forgot it um I usually refer to this great paper or rather essay by Carl priston namely beyond the desert landscape uh from uh the book Andy Clark and his critics in which uh he beautifully uh explains uh the difference between low road and high road and why active inference literature relies more recently on developing the high road approach instead of the low road one so I highly recommend reading uh that essay as well

Thomas yeah I mean kind of piggy back on that question on the high road I had a lot of interest in that I don't have experience knowing about Marco blankets I think that seems very clearly one of the major definitions of what we're getting at there so I can't make a comment about that yet I imagine we're going to talk about that in future chapters when I was reading that I reminded from Plato's Mino there there's discourse about the definition of figure and it's using a very particular mode of discussion about to talk about defining what color coming as a followup of figure so and it's a discussion of limit uh in fact even play's character has a

very particular kind of nuance
discussion that that can even challenge
the discussion what limit is so that's
is more just meditating on I think
that's very I'm interested in hearing
more about that um at this
point yeah we'll definitely come to uh
markup blank get in chapter three and uh
hopefully we'll have an extensive
discussion around that uh essential
concept and I agree markup blanket um I
mean holds Central um central place in
all active inference literature so
uh thanks Daniel

Jeff yeah I'm just I'm just reminded uh
not from the last bit of conversation
but just a little bit before that and
I'm sorry I'm jumping in in the middle
here uh but I'm just reminded of
wheeler's partici for a Universe
[Music]

um wheeler
participant okay so uh interesting uh
you see uh I'm not uh very familiar with
Wheeler's work but um as far as I know
uh I mean some of the ideas from um I
mean uh some of Wheeler's ideas and
specifically uh
the uh bestowing a kind of
fundamentality to the concept of
information uh which is one of the um
wheelers as far as I know um main
arguments uh so uh
is very very similar to what basian
mechanics um way of modeling uh the
physics and and I don't mean just physic
of belief but actually physics of
everything so I'm not sure if you've

heard

B Bayesian mechanics but it's a kind of new way of uh doing uh even fundamental physics not only uh as a way to model cognitive agents uh but U it is much more fundamental than that uh so uh as the work the works of people like um Delton activ Adel would show uh this new kind of approach to doing uh physics is uh in some sense perfectly congruent with everything uh that has come so far in in the history of fundamental physics in terms of its classical I mean classical physics quantum physics and even for modeling uh dynamical chaotic systems so

uh yes um the uh the notion of uh fundamentality of information or rather the fundamentality of beliefs belief in its statistical sense uh which is a kind of uh a mod modulated information is a kind of uh is a kind of information that has a specific probability

distribution uh is Central to um I mean uh almost everything active inference related or Fe related so yes I think uh they're closely related to Wheeler's work as well and also obviously Chris Fields a great course on in physics as information processing

uh has some great discussions around that point as

well I'm editing the journal right now yeah Jeff that will be awesome to see some connections with Chris fields and and to make a comment on this I think cognitive science in a way is looking for that kind of

qualified description of things in their
environment figure and ground and agent
in environment ecological
psychology and
um there are various like
seemingly uh overly simplistic to the
point of losing their epistemic or
pragmatic value theories like um
theories that might be criticized as um
like for example even in this discussion
words that have been brought up like
mentalizing or
anthropocentric so clearly those could
be taken to a level where it's like yeah
it we you know out mentalized where it's
not even useful and it's anthropocentric
where it doesn't even apply to this or
that um or the other way around and then
also anything that's part of our
behavior and experience it is coming in
some way through the apparatus of the
person so it can only be escaped so
fundamentally and so sometimes there's a
movement away from so-called
anthropocentrism but then it's like if
it doesn't have us doing that theory
then what is that really even trying to
describe
then David
yeah hi good morning yes I just wanted
to uh see if uh the mous Utopia studies
would fall into this dynamic as well uh
performed by John Calhoun in the 60s I'm
not sure if you're familiar with the
studies
themselves in terms of uh the behavioral
modeling um because they do describe it
as a collapse in behavior and in terms

of a
predetermined you know Utopia from a
different
experiential point of view and then
seeing how that deteriorated despite the
constraints on the model itself um
wanted to know how that might speak
towards uh you know the behavioral
modeling itself and uh the collapse
function how that might be related to
the basium mechanics as well in terms of
uh defining an environmental space you
know for another U active group of
agents for
example thanks David I'm not familiar
with this work at all so thanks for
bringing
it up and introducing it um I don't know
if anyone is familiar with mous Utopia
experiments and
uh yeah Jeff go for
it no I I that that leads directly into
I think um Mar OB Blake and in the
holographic principle um and and um
layers of separation between agents and
the translation that has to occur at
every hop right right um for example
we're not going to be able to empathize
with what it's like to be a bacteria in
a bacterial Colony but for the
bacteria they take a very bacterio
Centric view of the world so I think
it's very natural for us and it's right
for us to take a anthropocentric uh
approach right because that's our lived
experience and I think there's definite
overlap with with with with psychology
and soci the social sciences in general

there right um we can't imagine what it's like to be a sun but if you look at tile hard's complexity Theory right like anything with sufficient complexity has Consciousness so you know the sun knows what it's like to be a sun the Sun but but but you know it's building that it's building that translators and the steps between the different systems is is really what what I'm interested in getting after right so it's kind of like the rock being able to imagine it's a rock but not interact with another rock for example yeah yeah yeah exactly yeah I just found it interesting because the uh the mouse St itself where they they teered the different areas and then they kind of segregated it so you know they had you know different levels but you know that's also coming from my background which is sociology so I mean there's a little bit of Prior from myself there as well so so I'd also have to learn more about the experiment specifically but I think one phenomena it speaks to is the unified nature of cognitive phenomena in the developmental in The evolutionary setting like you could study just working memory and you go down the rabbit hole and then working memory is going to bleed off into some other cognitive phenomena because it's all happening in part of this integrated unit of of organismal function and then when we take that view out to like the intergenerational and the social so they're like vertical and horizontal

transmission all these different kinds
of inheritance mechanisms then
oh let's just invent a pencil to offload
this and then there's nonlinear effects
or let's just make this well make this
thing that was clear now let's paint
over that like changes to what what can
um influence one cognitive phenomena or
one kind of intervention or change in
the game can lead to like these
cascading effects all these different
kinds of ecological

Dynamics so that kind of motivates that
kind of motivates at least the
integrated framework to prepare for the
um proper analysis of these complex
situations

yeah do you think those decision non
self motivated in that sense you know
where um you know something that's
single cellular might you know has two
fella for example it's going to drop one
um because it may not you know be able
to exist with two in that in that you
know environment itself but is that
necessarily a choice or is that a
product of the environment
itself and also uh I mean the whole
verging field of Basil cognition also
deals uh with these kinds of questions I
mean whether uh those single celled
organisms uh actually possess kind of uh
primitive U cognition uh in terms of
memory learning decision making and so
on and uh how it how it's implemented
biologically so uh one uh strong
candidate for um for the way I mean for
uh the

implementation of uh the basil cognition
in uh any uh even the single cell
organisms uh is through the uh Gap
Junction distribution of their of their
bioelectric network uh which is the main
uh research one of the
main areas of research um people such as
Michael Levan and his colleagues uh so
yeah it's a very interesting um
developments in recent uh biology uh
so the whole are of Basil cognition as
far as I know uh officially started from
2021 so they're very new questions and
uh obviously with the mainstream uh
biology concerned uh concerned on uh uh
the modern synthesis uh approach uh it's
a long way uh to somehow nudge uh nudge
the Paradigm and uh to make a through
true paradigm shift in biology but uh
it's happening and uh I I believe uh
they can lead to uh some really exciting
uh new work in this area once uh we
escape uh the traditional uh Gene
centered uh view of the organisms uh to
the the more holistic and organism
centered um ontology so that can that
can also apply
to uh even non-biological agents if we
want to uh think of the Continuum
between uh biological things and
non-biological ones as a true Continuum
not not not uh something that
uh that something that has any kind of
um
fundamental uh difference between
between them so basian mechanics uh is
one of the approaches that somehow
dismantles uh this um discrepant this

fundamental discrepancy between uh
biological and
non-biological agents uh so yeah I
believe uh it's a much more interesting
worldview to look at things this way in
a kind of Continuum as a kind of process
and so on instead of uh aristan
essentialism that's great I'll just add
one more thing I don't want to give too
much input but you jogged my memory on
something related to something I was
thinking about with uh Len's work as
well in terms of you know if you have a
more macro level like a human being for
example and you have uh you know an
accident or something and a limb needs
to be removed we have this you know
Phantom limb we call it where the
individual still feels like they can
move the limb the limb might still be
there but you know it has been removed
so obviously there is some memory there
but we also know that we have neurons
you know and extremities in our bodies
you know to what extent is that you know
a mechanic of just cognition or is it
also a mechanic of the potential energy
that was lost in that space for example
and would we be able to pinpoint that
kind of thing with the technology we
have now to try and rep replicate that
on a smaller scale to kind of find that
you
know bioelectric signal or whatever else
might be going on in terms of you know
the interconnectivity between the micro
and the macro and that
Dynamic thank you great uh okay Cohen I

noticed you had your hand raised if I'm not mistaken you want to add anything or

or

ywn if I had my hand raised that was a mistake sorry but I I yesterday I listened to the fron and and Levin talk and uh it it seemed super interesting to me that Levin thinks that like

he has simulated some systems where like some genotypes understand that other agents share it the genotype and uh they Fair much better and they like communicate much better with agents that are very similar themselves yeah

interesting uh actually we'll have a lot more to talk about uh all of these things in the coming chapters and uh as I said it's uh one of the uh main focuses of uh active inference uh especially uh from its high level approach so um I'm excited to hear all the I mean your opinions on all of these matters uh as we come uh to those points as well

so Jeff uh did you want to add anything or yeah um I my background is National Security military and uh intelligence community and working overseas like for for the US Military and um there's a massive problem right now with security classification and also with also Enterprise it and cyber security um massive problem with with um information classification controlling classified information and

um zero trust and protecting information

on the internet right the internet is a
scale-free system uh so I think there's
a practical applicability here to
fundamentally redesigning the way that
we look at protecting our information on
the internet and this really could be a
practicum and I've got about 6,000 words
uh in a paper if anybody's interested in
collaborating uh short deadline uh I'm
targeting it for a conference called the
hammeron which is um one of the big
National Security conferences in the
country it's right outside the NSA
headquarters and um yeah so I'm I'm I'm
actively working on that um and I I
think it's a very very um like re like
Salient point and And Timely Point um to
our current
conversation very
interesting thanks for bringing it up if
anyone is interested in
collaboration
uh that would be exciting and
great okay so quick question sorry aliot
what would be the best way through this
group uh to be in touch about things
like that in the future I'm not sure Jee
if if uh my background would be would be
especially helpful on that but I'm just
curious about like what's the the sort
of best practice on how to link up for
that kind of thing if
so I'll just answer it um on the cooh
six page you can add like a row with
your contact
info you could also message someone in
Zoom if you're synchronously here or
just find any other way another option

is to write on the
projects
page and that can include your contact
information so if you want to kind of
check out other people's projects or
plant your own um here if you want to
kind of just do it more personally at
the this page also understanding that
everything that's in the textbook group
document is like open essentially and
just contact each other and enjoy beyond
the halls of The

Institute awesome

thanks I'm like Curious just any anyone
else um even if they haven't added
anything like just how is chapter
one how did they feel like it opened the
book or what did it leave them
with yes

Petra yeah uh hi I have a question about
uh uh normative framework

um I'm just thinking is it like uh the
framework describes uh healthy
functioning organism when we are looking
at a human uh as a as a agent
or uh is it meant uh differently I'm
thinking um in a context of uh

Psychopathology when we are talking that
the uh agents model generative model is
somehow sub optimized uh whether it is
connected with this uh normative uh
aspect of framework does it make sense
sorry great question great question
thank you uh so um I'll take a first
step so uh uh you see uh the use of this
word normative here u in active
inference literature and also in the
textbook is uh somewhat uh I think

defined in the notes if I'm yes it was
in page U on page
267 so here it says the term normative
means uh there is some evaluative
standard against which behavior can be
scored active inference is normative in
the sense that perception and action are
scored by free energy a quantity we will
unpack throughout this and the next
chapters so as I said again uh it's more
technical uh sense of the word normative
and it's not necessarily congruent with
um other um connotations or meanings of
this word so uh basically uh the reason
uh it's I mean the word normative used
uh in active inference is to refer to uh
this Central parameter namely free
energy as the main optimization
parameter uh around which all the other
formulation and um I mean mathematical
formalism revolves so uh that's
basically uh as far as I understand uh
the meaning of the word the adjective
normative in active inference literature
but I'd be happy to hear any other
opinion uh on that matter as
well thanks Ali thank you Petra um I'll
just add a few other points especially
in the social and human there was some
great discussion on this question of
normativity in the social sciences
course with ma alasen um and Ben White
and others and AVL last year um so
especially in the and um it it is um
an interesting question and one that
also people have specifically looked at
in the um Psychopathology sense like
Carl friston and Thomas par two of the

three authors on the textbook are clinicians and so there's a lot of discussion about like well what does it mean if it's Bas optimal but it's DSM diagnosed like how how is it what is it that's going right or wrong what's optimal or nonoptimal and just different ways to talk about like what is optimal like under this um memory parameter this Behavior can be understood as optimal and so that's a kind of like narrow Bas optimal but then that's just what prepares for the discussion like but then the context this is not aligning with their preferences so it's a great question though and and also like um just the is and the art and oh well active inference can be applied to um this system you know we know it can be applied to any system oh can be applied to a company so it ought to be this way but a framework that just describes how things are how do you go from that to how it ought to be and also again as a side note uh I will refer to Mark bier's work as well because uh much of his research uh is um is concentrated on uh again the development of what it actually means uh for U for an organism or an agent to have a normative perception or normative action and uh what it what it means U is a kind of gold Direct edness action or kind of intentional intentional action or as what Terence Deacon would call intentional not necessarily intentional

uh so um this kind of normative uh
perception or normative action as uh
bicker develops in U many of his papers
U is the sense that uh this kind of
intentionality is uh unseparable from
the perception and obviously the action
but um
in in the so-called um tiop phobic uh
community of biologists uh this this
kind of discourse uh somehow sometimes
would uh raise eyebrows because uh as we
know uh from I don't know around U 1950
on word uh there's this kind of phobia
around this Theology and any kind of uh
I mean go directedness and U to put
intentionality uh and mechanistic
explanations and so on but uh as Terence
ston also shows an incomplete nature um
uh and as I said Mark bier's work um
this intentionality and
normativity uh are essential aspects of
uh any sentient and cognitive agent uh
and by uh agent I mean anything from
single cell organism to uh I don't know
much more complex on so
um uh thanks for your question yeah oh
yeah think great points Allan to kind of
connect the like what you described
earlier about the Theology and TI tiop
phobia and the Deacon and the absenti
nature the idea that there are systems
um for whom deterministic accounting of
their connections accounts for their
behavior like a a pendulum or something
like that but even a system um like a
pendulum could be described as like this
absenti movement towards and so there
are fundamental parts of especially

symbolic knowledge based systems that have um a teely logical cause like the cause of the house is that there was a desire to build a house in that place um and

so active inference especially in this chapter one setting it just like brings it all out on the table the chapter does and and the discussions do and um there's a lot in the chapter and also a lot that's kind of like a adjacency that's always fun to bring out and this is kind of just like calibrating to the scope of the kinds of things that that are our starting point for continuing with the journey and just seeing about like some of the heris stics and tools and frames that have been brought um up previously and then in this first half of the textbook we kind of like get through one worked example with the mamalian nervous system by chapter five of like this is kind of of the sketch of what it looks like to start applying this to a system but here's how much work it takes to get to this stage and here's what the model does and doesn't say at that

stage thanks

Daniel okay so at our final minutes um I'd be interested to hear uh now that you've read chapter one uh what do you anticipate uh or what do you hope uh to gain from Reading um the textbook all through I mean does reading chapter one change your anticipation for better or worse yes

as

yeah I think

Thomas raised his

hand oh uh sorry I have some problems

seeing the race okay Thomas thanks oh

sure um no it hasn't this chapter didn't

really change my anticipations mainly

because Daniel was so generous when I

talked with him in the Discord chat he

sent me that paper on order and change

in this art basically how to give the

active inference account could apply to

aesthetic experience phenomenal I mean

my research and writings for Plato's

account of Aesthetics like lines up

perfectly to what that paper was talking

about and if anything I'm just more in

I'm excited to move forward I think the

idea that we I think the account really

interesting um I think the fact that

priston does a great job making an

argument that aesthetic experience is

not being suppressed are kind of De

decomposed down um basically on the

account of epistemic arcs I think that's

outstanding um as an account for the

aesthetic

experience and I think the fact that

this can be rigorously discussed is

that's the area why I'm even here in the

first place because I TR I for anyone

doesn't remember call it see meaning me

I'm a lecturer in visual art at a

university and the biggest challenge I

have with pupil students is to

rigorously demonstrate how do we know

something's more successful or not so

this kind of normative measure measure

kind of approach um the fact that that
can be demonstrated in such rigorous
ways I'm very interested um but also the
mystery of basically hyra has a whole
paper on the question of what's the
thing this whole question of That's So
the kind of high road even question of
how to answer what is a thing uh is just
I'm very fascinated nothing's really
been changed I'm more just I'm enjoying
the meal I'm looking forward to the next
course that be my summary so
far thanks uh and on the point of
Aesthetics uh actually there's this
recent um exciting uh special issue um
and I think it was in um philosophical
transactions of Royal Society on the
subject of Aesthetics from the Viewpoint
of predictive
processing uh which
um okay yes AI exactly it was in this
issue also tomorrow um with uh Karen
Konan the guest stream um about
literature and predictive processing
that should be super exciting too I'll
put this in the notes
yeah she uses um the predictive
processing uh as
applied she applies predictive
processing uh to uh literary studies and
uh somehow uses literature as a kind of
experimental hub for investigating many
facets of Consciousness from the
Viewpoint of predictive processing it's
really fascinating
work awesome cool all right well thank
you Ali for the facilitation and thank
you everyone um next week we'll continue

to discuss chapter one in the later time
period or feel free to comment this time
again and we'll be um talking about
chapter 6 the recipe for generative
modeling so thank you all farewell
thanks for joining everyone bye
bye